

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: Aluminum cobalt isopropoxide, 10% w/v in isopropanol  
 Product Description: Aluminum cobalt isopropoxide, 10% w/v in isopropanol  
 Cat No. : 36593  
 Molecular Formula  $\text{Co}[\text{Al}(\text{OCH}(\text{CH}_3)_2)_4]_2$

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Flammable liquids                                  | Category 2 (H225) |
| Skin Corrosion/Irritation                          | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                  | Category 1 (H318) |
| Specific target organ toxicity - (single exposure) | Category 3 (H336) |

**Label Elements**


Signal Word

Danger

Hazard Statements

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H225 - Highly flammable liquid and vapor  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H336 - May cause drowsiness or dizziness

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                    | CAS No     | Weight % |
|------------------------------|------------|----------|
| Isopropyl alcohol            | 67-63-0    | 90.00    |
| Aluminum cobalt isopropoxide | 70504-56-2 | 10.00    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to

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protect themselves and prevent spread of contamination.

## Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes eye burns. Causes severe eye damage. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

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Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

## Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component         | Malaysia | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          |
|-------------------|----------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol |          | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> |

| Component         | European Union | The United Kingdom                                                                                                  | Germany                                                                                                                                                                                                                                                         |
|-------------------|----------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol |                | STEL: 500 ppm 15 min<br>STEL: 1250 mg/m <sup>3</sup> 15 min<br>TWA: 400 ppm 8 hr<br>TWA: 999 mg/m <sup>3</sup> 8 hr | TWA: 200 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 500 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 200 ppm (8 Stunden). MAK<br>TWA: 500 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 400 ppm<br>Höhepunkt: 1000 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and

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vapours filter Type A Brown conforming to EN14387  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                         |                                                                        |                                             |
|-----------------------------------------|------------------------------------------------------------------------|---------------------------------------------|
| Appearance                              | Violet                                                                 |                                             |
| Physical State                          | Liquid Solution                                                        |                                             |
| Odor                                    | Alcohol                                                                |                                             |
| Odor Threshold                          | No data available                                                      |                                             |
| pH                                      | No information available                                               |                                             |
| Melting Point/Range                     | No data available                                                      |                                             |
| Softening Point                         | No data available                                                      |                                             |
| Boiling Point/Range                     | No information available                                               |                                             |
| Flash Point                             | 12 °C / 53.6 °F                                                        | Method - No information available           |
| Evaporation Rate                        | No data available                                                      |                                             |
| Flammability (solid,gas)                | Not applicable                                                         | Liquid                                      |
| Explosion Limits                        | No data available                                                      |                                             |
| Vapor Pressure                          | No data available                                                      |                                             |
| Vapor Density                           | No data available                                                      | (Air = 1.0)                                 |
| Specific Gravity / Density              | No data available                                                      |                                             |
| Bulk Density                            | Not applicable                                                         | Liquid                                      |
| Water Solubility                        | No information available                                               |                                             |
| Solubility in other solvents            | No information available                                               |                                             |
| Partition Coefficient (n-octanol/water) |                                                                        |                                             |
| Component                               | log Pow                                                                |                                             |
| Isopropyl alcohol                       | 0.05                                                                   |                                             |
| Autoignition Temperature                | No data available                                                      |                                             |
| Decomposition Temperature               | No data available                                                      |                                             |
| Viscosity                               | No data available                                                      |                                             |
| Explosive Properties                    |                                                                        | Vapors may form explosive mixtures with air |
| Oxidizing Properties                    | No information available                                               |                                             |
| Molecular Formula                       | Co[Al(OCH(CH <sub>3</sub> ) <sub>2</sub> ) <sub>4</sub> ] <sub>2</sub> |                                             |
| Molecular Weight                        | 585.15                                                                 |                                             |

## SECTION 10: STABILITY AND REACTIVITY

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## Reactivity

None known, based on information available.

## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

No information available.  
None under normal processing.

## Conditions to Avoid

Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

None known.

## Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

##### **(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component         | LD50 Oral                                  | LD50 Dermal         | LC50 Inhalation       |
|-------------------|--------------------------------------------|---------------------|-----------------------|
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat ) | 72.6 mg/L ( Rat ) 4 h |

##### **(b) skin corrosion/irritation;**

Category 2

##### **(c) serious eye damage/irritation;**

Category 1

##### **(d) respiratory or skin sensitization;**

**Respiratory**

No data available

**Skin**

No data available

##### **(e) germ cell mutagenicity;**

No data available

##### **(f) carcinogenicity;**

No data available

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There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3

Results / Target organs Central nervous system (CNS), Respiratory system.

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component         | Freshwater Fish                                                                                                                                                                                              | Water Flea                                      | Freshwater Algae                                                                                     | Microtox                                           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Isopropyl alcohol | LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h (Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h (Daphnia) | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h | EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus) | = 35390 mg/L EC50 Photobacterium phosphoreum 5 min |

Persistence and degradability  
Persistence No information available  
Persistence is unlikely.

Bioaccumulative potential Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Isopropyl alcohol | 0.05    | No data available             |

Mobility in soil No information available.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

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## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

#### **Other Information**

Waste codes should be assigned by the user based on the application for which the product was used Do not flush to sewer Can be landfilled or incinerated, when in compliance with local regulations Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1219  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ISOPROPANOL

### Road and Rail Transport

UN-No UN1219  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ISOPROPANOL

### IATA

UN-No UN1219  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ISOPROPANOL

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component         | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Isopropyl alcohol | 200-661-7 | X    | X   | X     | X    | X    | X     | X    | KE-29363 |

| Component         | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Major<br>Accident Notification | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Safety<br>Report Requirements | Rotterdam Convention<br>(PIC) | Basel Convention<br>(Hazardous Waste) |
|-------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Isopropyl alcohol |                                                                                                    |                                                                                                   |                               | Annex I - Y42                         |

### National Regulations



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**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

30-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**